
Bayes-Optimal Knowledge Arbitration Under Conflict: Appendix Proofs

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Abstract

This appendix gives NeurIPS-style proof statements for Bayes-optimal knowledge arbitration under conflict. The main change relative to the informal theorem sketch is mathematical: the geometric arbitration rule is not Bayes-optimal for an arbitrary latent-reliability mixture model, but it is exactly optimal for a log-score/KL-compromise decision model with posterior mean reliability. The appendix also gives a minimax lower bound for fixed trust weights, and restructures the chain-of-thought calibration theorem so that overconfidence is derived from primitive conditions—endogenous reasoning traces, self-confirming likelihood separation, a sharp RLVR prior/effective learning rate, and misspecification-dependent critical learning rates—rather than assumed directly. A final corollary formalizes answer-token-only eta-tempered decoding as the inference-time repair suggested by the theory. The revised version also makes the CoT result stability-stratified and benchmark-conditional, and adds concrete empirical audits for answer stability and self-confirming likelihood separation.

1 Proof roadmap and submission notes

NeurIPS formatting. This source is written as a single-column NeurIPS-style appendix: 10pt body type, 5.5in text width, 9in text height, compact theorem blocks, natbib-compatible citations, and booktabs-style tables. The file includes a small local `neurips_2026.sty` fallback so that the PDF is reproducible in this environment. For an actual NeurIPS submission, replace that fallback with the official style file from the author kit and compile the same body text without changing margins or font sizes.

Mathematical safety. The proof package makes three conservative choices.

1. The geometric pool is proved as the solution of an explicit KL-compromise problem, not asserted to follow from every possible reliability likelihood.
2. The fixed-weight lower bound is proved by a separated binary-logit construction, so the constant-regret claim is not dependent on asymptotics or an informal adversary.
3. The chain-of-thought theorem uses primitive, testable conditions. The key reinforcement lemma derives monotone confidence growth from endogenous evidence and likelihood separation, while the critical-rate comparison derives conflict amplification from a larger misspecification gap.
4. The CoT statement is stability-stratified and benchmark-conditional: answer changes enter through an explicit flip-rate correction, and ConflictBank-style entity swaps and WikiContradict-style natural contradictions may have different separation parameters, critical rates, and recovery thresholds.

Table 1: Proof status and role of each theorem. The statements are phrased to avoid overclaiming beyond their assumptions.

Result	What is proved	Main proof device
Theorem 1	The reliability-weighted geometric pool is optimal under a log-score/KL-compromise arbitration model.	Strict convexity on the simplex and a Lipschitz perturbation bound for log-pools.
Theorem 2	Any fixed trust weight can suffer constant excess risk on a class containing temporal-update and misinformation conflicts.	Binary logit margins, plus a Le Cam/Pinsker no-signal lower bound.
Theorem 3	RLVR-style extended reasoning can produce conflict-amplified overconfidence under primitive endogenous-evidence and misspecified-Bayes conditions, with an explicit answer-flip correction.	Berk–Nash concentration, self-confirming likelihood separation, answer-stability stratification, and a benchmark-conditional critical-rate comparison.
Corollary 3.4	Answer-token eta-tempering reduces overconfidence when the raw effective learning rate exceeds the critical rate.	One-dimensional generalized-Bayes calibration model.

2 Shared setup

Let $X = (q, c)$ denote a factual query and retrieved context. The answer space $\mathcal{Y}$ is finite. Two predictive distributions are available:

$$P_X(y) = p_\theta(y | q), \quad Q_X(y) = p_{\text{ctx}}(y | q, c), \quad y \in \mathcal{Y}.$$

The parametric source P_X is the closed-book distribution; the contextual source Q_X is the context-conditioned distribution. We assume strict positivity: for some $\varepsilon > 0$,

$$P_X(y) \geq \varepsilon, \quad Q_X(y) \geq \varepsilon, \quad \forall X, y.$$

This is exactly what is obtained after standard probability smoothing. The lower bound can be weakened to positivity plus finite second moment of the log-likelihood-ratio oscillation introduced below.

Let $R \in [0, 1]$ be the latent reliability of the context. The posterior mean reliability is

$$w^*(X) = \mathbb{E}[R | X].$$

When the query is fixed and suppressed, Bayes’ rule gives

$$w^*(q, c) = \frac{\int_0^1 r p(c | q, r) \pi(r | q) dr}{\int_0^1 p(c | q, r) \pi(r | q) dr}.$$

For $w \in [0, 1]$, define the geometric aggregate

$$G_w^X(y) = \frac{P_X(y)^{1-w} Q_X(y)^w}{Z_X(w)}, \quad Z_X(w) = \sum_{z \in \mathcal{Y}} P_X(z)^{1-w} Q_X(z)^w. \quad (1)$$

The corresponding prediction is

$$\hat{y}_w(X) = \arg \max_{y \in \mathcal{Y}} G_w^X(y),$$

with deterministic tie-breaking. The 0–1 risk of a policy $\hat{y}$ is

$$R(\hat{y}) = \mathbb{E}[\mathbf{1}\{\hat{y}(X) \neq Y^*\}].$$

3 Theorem 1: Bayes-optimal log-pool arbitration

The first theorem is intentionally stated under a log-pool decision model. Without this assumption, a generic latent reliability model usually induces a linear posterior predictive mixture rather than a geometric pool.

Assumption 1 (Log-pool decision model). *Conditional on X and reliability $R = r$, the decision maker chooses a predictive distribution $p \in \Delta(\mathcal{Y})$ by minimizing*

$$\mathcal{L}_X(p, r) = (1 - r) \text{KL}(p \| P_X) + r \text{KL}(p \| Q_X). \quad (2)$$

Equivalently, the decision maker uses logarithmic-score utility and seeks the reliability-weighted information-geometric compromise between the parametric and contextual predictive experts.

Theorem 1 (Bayes-optimal log-pool arbitration). *Under the shared setup and [Assumption 1](#), the posterior-optimal aggregate is*

$$p^*(y | q, c) = G_{w^*(q, c)}^{(q, c)}(y) = \frac{p_\theta(y | q)^{1-w^*(q, c)} p_{\text{ctx}}(y | q, c)^{w^*(q, c)}}{\sum_{z \in \mathcal{Y}} p_\theta(z | q)^{1-w^*(q, c)} p_{\text{ctx}}(z | q, c)^{w^*(q, c)}}. \quad (3)$$

The Bayes action under 0–1 loss for this posterior predictive distribution is

$$\hat{y}^*(q, c) = \arg \max_{y \in \mathcal{Y}} p^*(y | q, c).$$

Moreover, for any plug-in estimate $\hat{w}(X)$,

$$R(\hat{y}_{\hat{w}}) - R(\hat{y}_{w^*}) \leq (\mathbb{E}[B_X^2])^{1/2} (\mathbb{E}[(\hat{w}(X) - w^*(X))^2])^{1/2}, \quad (4)$$

where

$$B_X = \text{osc}_{y \in \mathcal{Y}} \log \frac{Q_X(y)}{P_X(y)} = \max_y \log \frac{Q_X(y)}{P_X(y)} - \min_y \log \frac{Q_X(y)}{P_X(y)}. \quad (5)$$

In particular, if $P_X(y), Q_X(y) \geq \varepsilon$ for all X, y , then

$$R(\hat{y}_{\hat{w}}) - R(\hat{y}_{w^*}) \leq 2 \log(1/\varepsilon) (\mathbb{E}[(\hat{w}(X) - w^*(X))^2])^{1/2}.$$

Consequently, if a calibration procedure using signals $s(X)$ satisfies

$$\mathbb{E}[(\hat{w}(s(X)) - w^*(X))^2] = O(n^{-1}),$$

then the plug-in arbitration rule has excess risk $O(n^{-1/2})$.

Proof. Fix X . Taking the posterior expectation of (2) over $R | X$ gives

$$\mathbb{E}[\mathcal{L}_X(p, R) | X] = (1 - w^*(X)) \text{KL}(p \| P_X) + w^*(X) \text{KL}(p \| Q_X),$$

by linearity. Write $w = w^*(X)$, $P = P_X$, and $Q = Q_X$. The objective is

$$F(p) = \sum_{y \in \mathcal{Y}} p(y) \{ \log p(y) - (1 - w) \log P(y) - w \log Q(y) \}.$$

The Lagrangian for $\sum_y p(y) = 1$ is

$$\mathcal{J}(p, \lambda) = F(p) + \lambda \left(\sum_y p(y) - 1 \right).$$

The first-order condition is

$$\frac{\partial \mathcal{J}}{\partial p(y)} = \log p(y) + 1 - (1 - w) \log P(y) - w \log Q(y) + \lambda = 0,$$

so

$$p(y) = \exp(-1 - \lambda) P(y)^{1-w} Q(y)^w.$$

Normalization over $\mathcal{Y}$ yields $p = G_w^X$. Since F is strictly convex on the simplex whenever P and Q are positive, this minimizer is unique.

Given posterior predictive distribution G_w^X , the conditional 0–1 risk of predicting $a \in \mathcal{Y}$ is

$$\mathbb{P}(Y^* \neq a \mid X) = 1 - G_w^X(a),$$

which is minimized by any maximizer of $G_w^X(a)$. This proves the Bayes action statement.

For the excess-risk bound, let $G = G_{w^*(X)}^X$, $\hat{G} = G_{\hat{w}(X)}^X$, $a^* = \arg \max_y G(y)$, and $\hat{a} = \arg \max_y \hat{G}(y)$. Conditional on X ,

$$R_X(\hat{a}) - R_X(a^*) = G(a^*) - G(\hat{a}).$$

Since $\hat{G}(\hat{a}) \geq \hat{G}(a^*)$,

$$G(a^*) - G(\hat{a}) \leq |G(a^*) - \hat{G}(a^*)| + |G(\hat{a}) - \hat{G}(\hat{a})| \leq \|G - \hat{G}\|_1.$$

Set

$$A_X(y) = \log \frac{Q_X(y)}{P_X(y)}, \quad \bar{A}_w = \sum_y G_w^X(y) A_X(y).$$

Differentiating (1) gives

$$\frac{d}{dw} G_w^X(y) = G_w^X(y) (A_X(y) - \bar{A}_w).$$

Therefore

$$\left\| \frac{d}{dw} G_w^X \right\|_1 = \sum_y G_w^X(y) |A_X(y) - \bar{A}_w| \leq \text{osc}_y A_X(y) = B_X.$$

The mean-value theorem implies

$$\|G_{\hat{w}(X)}^X - G_{w^*(X)}^X\|_1 \leq B_X |\hat{w}(X) - w^*(X)|.$$

Taking expectations and applying Cauchy–Schwarz yields (4). If $P_X(y), Q_X(y) \geq \varepsilon$, then $Q_X(y)/P_X(y) \in [\varepsilon, 1/\varepsilon]$, hence $B_X \leq 2 \log(1/\varepsilon)$. The $O(n^{-1/2})$ claim follows immediately from the assumed mean-square convergence of $\hat{w}$. $\square$

Remark 1 (Why the original $\|P - Q\|_1$ bound belongs to a linear pool). For a linear aggregate $L_w = (1 - w)P_X + wQ_X$,

$$\|L_{\hat{w}} - L_w\|_1 = |\hat{w} - w| \|P_X - Q_X\|_1.$$

Thus a bound with $\|P_X - Q_X\|_1$ is exactly right for linear pooling. For geometric pooling, the correct Lipschitz factor is the log-likelihood-ratio oscillation B_X in (5).

Proposition 1 (When a JSD weight is a first-order reliability approximation). Let $T_X = \text{JSD}(P_X, Q_X)$. Suppose the reliability posterior satisfies a one-dimensional regular model

$$\log \frac{\mathbb{P}(R > 1/2 \mid T_X = t)}{\mathbb{P}(R \leq 1/2 \mid T_X = t)} = \alpha + \beta t + O(t^2)$$

near $t = 0$, and suppose the posterior for R is locally concentrated around its mode with observed information I_X . Then

$$w^*(X) = \mathbb{E}[R \mid T_X] = \sigma(\alpha) + \beta \sigma(\alpha)(1 - \sigma(\alpha))T_X + O(T_X^2) + O(I_X^{-1}),$$

where $\sigma(u) = (1 + e^{-u})^{-1}$. Hence a JSD-based weight is a first-order plug-in reliability estimator only under this explicit reliability-regression assumption; it is not a distribution-free consequence of Bayesian decision theory.

Proof. The Laplace approximation gives $\mathbb{E}[R \mid T_X = t] = \hat{r}(t) + O(I_X^{-1})$, where $\hat{r}(t)$ is the posterior mode. Under the stated local logistic specification, $\hat{r}(t) = \sigma(\alpha + \beta t + O(t^2))$. Taylor expansion of σ at α yields the displayed formula. $\square$

4 Theorem 2: fixed trust policies are minimax-suboptimal

For binary answers, write

$$a_X = \log \frac{P_X(1)}{P_X(0)}, \quad b_X = \log \frac{Q_X(1)}{Q_X(0)}.$$

The geometric rule predicts answer 1 exactly when

$$(1 - w)a_X + wb_X \geq 0. \quad (6)$$

Assumption 2 (Conflict realizability). *The benchmark class contains both temporal-update examples, on which the context-favored answer is correct, and misinformation examples, on which the context-favored answer is incorrect. More precisely, for every fixed $w_0 \in [0, 1]$, the class contains binary conflicts with logit margins satisfying the inequalities used in the construction below.*

Theorem 2 (Minimax lower bound for fixed arbitration). *Fix any constant $w_0 \in [0, 1]$. Under Assumption 2, there exists a distribution $\mathcal{P}_{w_0}$ over (X, Y^*) such that the fixed rule $\hat{y}_{w_0}$ has constant excess risk:*

$$R(\hat{y}_{w_0}; \mathcal{P}_{w_0}) - R(\hat{y}_{w^*}; \mathcal{P}_{w_0}) \geq \delta$$

for a numerical constant $\delta > 0$. In the deterministic construction below, $\delta = 1$. If the calibration signal $s(X)$ identifies the conflict type and a reliability estimator satisfies

$$\mathbb{E}[(\hat{w}(s(X)) - w^*(X))^2] = O(n^{-1}),$$

then

$$R(\hat{y}_{\hat{w}}; \mathcal{P}_{w_0}) - R(\hat{y}_{w^*}; \mathcal{P}_{w_0}) = O(n^{-1/2}).$$

With a positive decision margin, the same plug-in policy obtains exponentially small classification regret in n by Hoeffding concentration.

Proof. We construct two types of examples.

First consider $w_0 = 0$, the always-parametric rule. Let $Y^* = 1$, let $a_X = -A < 0$, and let $b_X = B > 0$. The parametric source favors 0 while the context favors the correct answer 1. The fixed rule with $w_0 = 0$ predicts 0 and is wrong. The adaptive oracle with $w^*(X) = 1$ predicts 1 and is correct.

Next consider $w_0 = 1$, the always-context rule. Let $Y^* = 0$, let $a_X = -A < 0$, and let $b_X = B > 0$. The parametric source favors the correct answer 0 while the context favors the wrong answer 1. The fixed rule with $w_0 = 1$ predicts 1 and is wrong. The adaptive oracle with $w^*(X) = 0$ predicts 0 and is correct.

Now fix $w_0 \in (0, 1)$. We build a distribution supported on two examples. On the temporal-update component T , set $Y^* = 1$, $a_T = -A_T < 0$, and $b_T = B_T > 0$. Choose

$$A_T > \frac{w_0}{1 - w_0} B_T.$$

Then

$$(1 - w_0)a_T + w_0b_T = -(1 - w_0)A_T + w_0B_T < 0,$$

so the fixed rule predicts 0 and is wrong, while the adaptive rule $w^*(T) = 1$ predicts 1 and is correct.

On the misinformation component M , set $Y^* = 0$, $a_M = -A_M < 0$, and $b_M = B_M > 0$. Choose

$$B_M > \frac{1 - w_0}{w_0} A_M.$$

Then

$$(1 - w_0)a_M + w_0b_M = -(1 - w_0)A_M + w_0B_M > 0,$$

so the fixed rule predicts 1 and is wrong, while the adaptive rule $w^*(M) = 0$ predicts 0 and is correct.

Let $\mathcal{P}_{w_0}$ put probability $1/2$ on each of T and M . The fixed rule is wrong with probability one and the adaptive oracle is correct with probability one. Hence the excess risk is 1. If stochastic labels or non-unit masses are used, the same construction yields any constant lower bound δ below the mass of the separated conflict components.

The plug-in convergence claim follows from [Theorem 1](#). If $s(X)$ identifies T versus M , then $w^*(s) = 1$ on T and $w^*(s) = 0$ on M . Estimating the conditional reliability in each stratum by the empirical mean gives mean-square error $O(n^{-1})$ when each stratum has probability bounded away from zero. Substitution into [Theorem 1](#) gives $O(n^{-1/2})$ excess risk. If the logit margins are bounded below by $\gamma > 0$, the plug-in rule changes its discrete prediction only if $|\hat{w} - w^*|$ exceeds a positive threshold; Hoeffding’s inequality then gives exponentially small classification regret. $\square$

Proposition 2 (No-signal lower bound). *Let $V \in \{0, 1\}$ denote whether an example is temporal-update or misinformation, with $\mathbb{P}(V = 0) = \mathbb{P}(V = 1) = 1/2$. Suppose the optimal action differs between the two values of V . If the observation X contains at most $I(V; X)$ nats of information about V , then every policy has error probability at least*

$$\frac{1}{2} \left(1 - \sqrt{2I(V; X)} \right)_+.$$

Consequently, if choosing the wrong component-specific action costs at least $\gamma > 0$, every policy has excess risk at least

$$\frac{\gamma}{2} \left(1 - \sqrt{2I(V; X)} \right)_+.$$

In particular, when $I(V; X) = 0$, no policy can achieve $o(1)$ excess risk.

Proof. For equal priors, the minimum probability of misclassifying V from X is

$$P_e^* = \frac{1}{2} (1 - \text{TV}(P_{X|V=0}, P_{X|V=1})).$$

Let $M = \frac{1}{2}P_{X|V=0} + \frac{1}{2}P_{X|V=1}$. The mutual information is the Jensen–Shannon divergence

$$I(V; X) = \frac{1}{2} \text{KL}(P_{X|0} \| M) + \frac{1}{2} \text{KL}(P_{X|1} \| M).$$

By Pinsker and the triangle inequality,

$$\text{TV}(P_{X|0}, P_{X|1}) \leq \text{TV}(P_{X|0}, M) + \text{TV}(P_{X|1}, M) \leq \sqrt{2I(V; X)}.$$

Substitution yields the error lower bound. If the wrong component-specific action costs at least γ , excess risk is at least γP_e^* . $\square$

5 Theorem 3: RLVR-conditioned calibration degradation under extended reasoning

The original informal theorem can sound circular if it assumes monotone reinforcement and then proves monotone overconfidence. This section avoids that by deriving the reinforcement inequality from a primitive property of self-generated reasoning traces: a trace sampled from the model’s current state is more likely under states that keep the same answer than under states that switch to a competing answer. The result is still conditional, as it must be; it is a theorem about a stylized generalized-Bayes model of chain-of-thought generation, not a universal law about all LLMs.

5.1 Primitive model

Let $\tau \in \{R, H\}$ denote training type, where R is RLVR/GRPO-style verifiable-reward training and H is helpfulness/RLHF/DPO-style training. Let s denote scale, b index the benchmark or conflict generator, and $c \in \{0, 1\}$ indicate whether the query-context pair contains conflict. A chain-of-thought trace consists of tokens $Z_1, Z_2, \dots, Z_K$. The model has an internal predictive state $\theta \in \Theta$, answer distribution $\mu_\theta \in \Delta(\mathcal{Y})$, and trace-token kernel m_θ . All CoT quantities below are conditional

on a fixed benchmark b ; for readability we suppress b in D_K , C_K , $\hat{Y}_K$, and OC_K , but we keep b in the regime parameters $\delta_{b,c}(s)$, $g_{b,c}(s)$, $\bar{\eta}_{b,c}(s)$, $K^*(b, c, s)$, and s_b^* . This benchmark index is necessary because artificial entity-swap conflicts and naturalistic contradictions can induce different self-confirming separations and recovery thresholds.

For any state θ , let

$$a(\theta) = \arg \max_{y \in \mathcal{Y}} \mu_\theta(y)$$

with deterministic tie-breaking, and define the answer basin

$$\mathcal{B}_a = \{\theta \in \Theta : a(\theta) = a\}.$$

The answer margin is

$$D(\theta) = \log \frac{\mu_\theta(a(\theta))}{\max_{y \neq a(\theta)} \mu_\theta(y)}, \quad C(\theta) = \sigma(D(\theta)).$$

After K tokens, write $D_K = D(\theta_K)$, $C_K = C(\theta_K)$, and $\hat{Y}_K = a(\theta_K)$. The signed overconfidence gap is

$$\text{OC}_K(c, s, \tau) = \mathbb{E}[C_K - \mathbf{1}\{\hat{Y}_K = Y^*\} \mid c, s, \tau]. \quad (7)$$

When the model is bin-wise overconfident, ECE is equal to this signed gap up to binning error; otherwise (7) is the clean quantity controlled by the theorem.

Assumption 3 (Endogenous generalized-Bayes trace update). *Conditional on the current state θ_k , the next reasoning token satisfies $Z_{k+1} \sim m_{\theta_k}$. The model update is represented by a generalized posterior*

$$\pi_{k+1}(\theta) \propto \pi_k(\theta) \exp\{-\eta_\tau(s)\ell(\theta, Z_{k+1})\}, \quad (8)$$

and θ_{k+1} is a deterministic summary of π_{k+1} , such as its mode, mean embedding, or local Laplace center. The effective learning rate $\eta_\tau(s) > 0$ summarizes both training-induced prior sharpness and inference-time update strength.

Assumption 4 (Self-confirming likelihood separation). *For each answer basin $\mathcal{B}_a$ and competing basin $\mathcal{B}_{a'}$, $a' \neq a$, there is a benchmark-conditional separation $\delta_{b,c}(s) > 0$ such that, for every $\theta \in \mathcal{B}_a$ in the pre-saturation region and every pair $\vartheta_a \in \mathcal{B}_a$, $\vartheta_{a'} \in \mathcal{B}_{a'}$ with comparable prior mass,*

$$\mathbb{E}_{Z \sim m_\theta} [\ell(\vartheta_{a'}, Z) - \ell(\vartheta_a, Z)] \geq \delta_{b,c}(s). \quad (9)$$

For a fixed benchmark b , conflict weakly increases this separation: $\delta_{b,1}(s) \geq \delta_{b,0}(s)$. Across benchmarks, no ordering is assumed. This is a likelihood-separation condition on self-generated traces; it does not mention calibration or overconfidence.

Assumption 5 (Sharp RLVR prior/effective rate). *For the regime of interest, RLVR-style training produces a larger effective update strength than helpfulness tuning:*

$$\eta_R(s) \geq \eta_H(s).$$

It also places larger prior odds on a single answer basin after an initial answer has been selected. Formally, for the current basin $\mathcal{B}_a$ and any competitor $\mathcal{B}_b$,

$$\log \frac{\pi_0(\mathcal{B}_a)}{\pi_0(\mathcal{B}_b)} \Big|_{R,s} \geq \log \frac{\pi_0(\mathcal{B}_a)}{\pi_0(\mathcal{B}_b)} \Big|_{H,s}.$$

This is a primitive training-type condition, not a conclusion of the theorem.

Assumption 6 (Conflict misspecification and critical rate). *Let $g_{b,c}(s)$ denote the misspecification gap between the truth-generating process and the family of self-conditioned trace laws reachable by the model at scale s on benchmark b under conflict condition c . For fixed b , conflict increases misspecification: $g_{b,1}(s) \geq g_{b,0}(s)$. The generalized-Bayes update has a critical learning-rate map $\bar{\eta}(g)$ that is nonincreasing in g , and we write*

$$\bar{\eta}_{b,c}(s) = \bar{\eta}(g_{b,c}(s)), \quad \bar{\eta}_{b,1}(s) \leq \bar{\eta}_{b,0}(s).$$

When $\eta_\tau(s) > \bar{\eta}_{b,c}(s)$, the generalized posterior can concentrate on a self-confirming KL projection rather than on a truth-representing state, in the sense of misspecified Bayesian concentration and generalized-Bayes hypercompression.

Assumption 7 (Correction channel). *After a model-family- and benchmark-dependent length $K^*(b, c, s)$, an exogenous or truth-grounded correction channel may produce tokens or state transitions that favor the correct answer basin by expected margin $\xi_{b,c}(s, D_K) \geq 0$. For $s \geq s_b^*$ under conflict on benchmark b , the correction channel is dominated by self-confirming reinforcement until saturation:*

$$\xi_{b,1}(s, D) \leq \eta_R(s)\delta_{b,1}(s) \quad \text{for } D < D_{\max}.$$

For $s < s_b^$ under conflict on benchmark b , once $D_K > D_{\text{rec}}$ the correction channel strictly dominates self-confirming reinforcement:*

$$\xi_{b,1}(s, D) - \eta_R(s)\delta_{b,1}(s) \geq \zeta > 0.$$

5.2 Endogenous concentration and reinforcement

Lemma 1 (Berk–Nash concentration for endogenous trace evidence). *Under Assumption 3, suppose that conditional on a limiting state θ , the trace process is ergodic with stationary law m_θ , and that a uniform law of large numbers holds for $\ell(\vartheta, Z)$ over $\vartheta \in \Theta$. Then every stable concentration point of (8) belongs to the Berk–Nash set*

$$\Theta_{\text{BN}} = \left\{ \theta : \theta \in \arg \min_{\vartheta \in \Theta} \mathbb{E}_{Z \sim m_\theta} \ell(\vartheta, Z) \right\}. \quad (10)$$

If the model is misspecified, Θ_{BN} need not contain a truth-representing state. Thus endogenous CoT evidence can concentrate belief on a self-confirming answer basin.

Proof. On any subsequence along which the predictive state converges to θ , ergodicity and the uniform law of large numbers imply

$$K^{-1} \sum_{k=1}^K \ell(\vartheta, Z_k) \rightarrow \mathbb{E}_{Z \sim m_\theta} \ell(\vartheta, Z)$$

uniformly in ϑ . The generalized posterior (8) places exponentially larger mass on minimizers of the limiting expected loss than on non-minimizers. Therefore any stable concentration point must minimize expected loss under the data distribution induced by its own predictions, which is exactly (10). Under misspecification, this fixed point is self-confirming rather than necessarily true. $\square$

Lemma 2 (Self-confirming trace likelihood increases answer-basin odds). *Under Assumptions 3 and 4, before saturation, the posterior log-odds of the current answer basin against any competing basin increase in expectation by at least $\eta_\tau(s)\delta_{b,c}(s)$ per reasoning token:*

$$\mathbb{E} \left[\log \frac{\pi_{k+1}(\mathcal{B}_a)}{\pi_{k+1}(\mathcal{B}_b)} - \log \frac{\pi_k(\mathcal{B}_a)}{\pi_k(\mathcal{B}_b)} \middle| \theta_k \in \mathcal{B}_a, b, c, s, \tau \right] \geq \eta_\tau(s)\delta_{b,c}(s) - o(1), \quad (11)$$

where the $o(1)$ term is the Laplace or within-basin approximation error.

Proof. For representative states $\vartheta_a \in \mathcal{B}_a$ and $\vartheta_b \in \mathcal{B}_b$, the generalized posterior update gives

$$\log \frac{\pi_{k+1}(\vartheta_a)}{\pi_{k+1}(\vartheta_b)} - \log \frac{\pi_k(\vartheta_a)}{\pi_k(\vartheta_b)} = \eta_\tau(s) \{ \ell(\vartheta_b, Z_{k+1}) - \ell(\vartheta_a, Z_{k+1}) \}.$$

Taking conditional expectation with $Z_{k+1} \sim m_{\theta_k}$ and using (9) yields the stated lower bound for point states. Integrating over each basin and applying a local Laplace approximation converts point-state odds to basin odds with only the displayed $o(1)$ error. $\square$

Lemma 3 (Critical-rate conflict comparison). *Under Assumption 6, fix a benchmark b . If $\eta_R(s) > \bar{\eta}_{b,1}(s)$, then RLVR reasoning is in the inconsistent or hypercompressive regime under conflict on b . If, in addition, $\eta_R(s) \leq \bar{\eta}_{b,0}(s)$, then the same model can be below the critical rate without conflict on b . More generally, since $\bar{\eta}_{b,1}(s) \leq \bar{\eta}_{b,0}(s)$, conflict weakly enlarges the set of scales and effective learning rates for which $\eta_R(s) > \bar{\eta}_{b,c}(s)$.*

Proof. This is immediate from the monotonicity of $\bar{\eta}(g)$ and $g_{b,1}(s) \geq g_{b,0}(s)$. The substantive content is the primitive critical-rate transport condition in [Assumption 6](#); the proof only uses its ordering consequence. $\square$

Lemma 4 (Answer-stability correction). *Fix a benchmark b , conflict condition c , scale s , and training type τ . For $K_1 < K_2$, define the endpoint flip probability*

$$F_{12}^{(b,c,s,\tau)} = \mathbb{P}(\hat{Y}_{K_2} \neq \hat{Y}_{K_1} \mid b, c, s, \tau).$$

Then

$$\text{OC}_{K_2} - \text{OC}_{K_1} \geq \mathbb{E}[C_{K_2} - C_{K_1} \mid b, c, s, \tau] - F_{12}^{(b,c,s,\tau)}. \quad (12)$$

More generally, if $\mathcal{K}$ is a finite set of CoT budgets and

$$S_{\mathcal{K}}^{(b,c,s,\tau)} = \mathbb{P}(\hat{Y}_K \text{ is constant over } K \in \mathcal{K} \mid b, c, s, \tau),$$

then any comparison between two budgets in $\mathcal{K}$ pays at most the non-stability mass $1 - S_{\mathcal{K}}^{(b,c,s,\tau)}$ as an accuracy-change correction.

Proof. Let $A_K = \mathbf{1}\{\hat{Y}_K = Y^*\}$. By definition,

$$\text{OC}_{K_2} - \text{OC}_{K_1} = \mathbb{E}[C_{K_2} - C_{K_1}] - \mathbb{E}[A_{K_2} - A_{K_1}].$$

On the event $\{\hat{Y}_{K_2} = \hat{Y}_{K_1}\}$ the true label is fixed, so $A_{K_2} = A_{K_1}$. Therefore $A_{K_2} - A_{K_1} \leq \mathbf{1}\{\hat{Y}_{K_2} \neq \hat{Y}_{K_1}\}$ pointwise. Taking conditional expectations gives (12). The finite-budget version follows by replacing the endpoint flip event with the union event that the predicted answer is not constant over $\mathcal{K}$. $\square$

5.3 Calibration degradation theorem

Theorem 3 (RLVR-conditioned CoT calibration degradation). *Assume [Assumptions 3 to 7](#). Fix a benchmark b . For RLVR-trained models in the pre-saturation range, the following hold.*

(a) **General overconfidence growth with answer-flip correction.** *For $K_1 < K_2 \leq K^*(b, c, s)$, let*

$$\Delta C_{12}^{(b,c,s,R)} = \mathbb{E}[C_{K_2} - C_{K_1} \mid b, c, s, R], \quad F_{12}^{(b,c,s,R)} = \mathbb{P}(\hat{Y}_{K_2} \neq \hat{Y}_{K_1} \mid b, c, s, R).$$

If the confidence map is monotone in posterior basin odds, then

$$\text{OC}_{K_2}(c, s, R) - \text{OC}_{K_1}(c, s, R) \geq \Delta C_{12}^{(b,c,s,R)} - F_{12}^{(b,c,s,R)} - o(1).$$

Consequently, overconfidence is nondecreasing whenever the endpoint flip rate is no larger than the confidence increment. On the stable stratum, $F_{12} = 0$, which recovers the simpler monotonicity statement. If the model is bin-wise overconfident over this range, the same statement holds for ECE up to binning error.

(b) **Conflict amplification.** *In the pre-saturation range,*

$$\mathbb{E}[\text{OC}_{K+1} - \text{OC}_K \mid b, c = 1, s, R] \geq \mathbb{E}[\text{OC}_{K+1} - \text{OC}_K \mid b, c = 0, s, R] - o(1),$$

provided the conflict-minus-no-conflict accuracy-change term is no larger than the extra confidence increment induced by $\delta_{b,1}(s) \geq \delta_{b,0}(s)$ and $\bar{\eta}_{b,1}(s) \leq \bar{\eta}_{b,0}(s)$.

(c) **Benchmark-conditional scale-dependent self-correction failure.** *Under conflict on benchmark b , if $s \geq s_b^*$ then D_K reaches or remains near the saturation boundary $D_{\max}$, and*

$$\text{OC}_{K_{\max}}(1, s, R) = \text{OC}_{K^*}(1, s, R) + o(1)$$

whenever $K_{\max}$ lies in the saturated regime. If $s < s_b^$ and the post- K^* window is long enough for the correction channel to operate, then*

$$\text{OC}_{K_{\max}}(1, s, R) < \text{OC}_{K^*}(1, s, R)$$

up to the accuracy-change term induced by correction.

(d) **Training-type dependence.** The conclusions are sufficient-condition statements. They apply to training families satisfying the RLVR sharpness, self-confirming separation, and critical-rate conditions. They are not implied for helpfulness-tuned models with $\eta_H(s) \leq \bar{\eta}_{b,1}(s)$ or without self-confirming likelihood separation.

Proof. By Lemma 2, the posterior log-odds of the current answer basin grow in expectation by at least $\eta_R(s)\delta_{b,c}(s) - o(1)$ before saturation. The confidence map $C_K = \sigma(D_K)$ is monotone in the margin and, by assumption, the margin is monotone in the same basin odds. Thus

$$\mathbb{E}[C_{K+1} - C_K \mid b, c, s, R] \geq 0 - o(1)$$

in the pre-saturation window. Summing from K_1 to $K_2 - 1$ yields the confidence increment $\Delta C_{12}^{(b,c,s,R)}$ up to the same approximation error. Applying Lemma 4 gives the displayed lower bound with the endpoint flip correction $F_{12}^{(b,c,s,R)}$. This is the non-circular version of the accuracy-stability condition: the theorem either applies exactly on the stable stratum or pays an explicitly measurable penalty for answer changes.

For part (b), Assumption 4 gives $\delta_{b,1}(s) \geq \delta_{b,0}(s)$, so the per-token log-odds increment from Lemma 2 is weakly larger under conflict on the same benchmark. Lemma 3 adds that conflict weakly lowers the benchmark-specific critical rate, making hypercompressive concentration easier under conflict. Since C_K is monotone in basin odds and saturation occurs at high confidence, the expected confidence increment under conflict is at least that without conflict up to $o(1)$. Subtracting the measured accuracy-change terms, or equivalently applying the flip-rate correction from Lemma 4 in both regimes, gives the displayed inequality under the stated dominance condition.

For part (c), after $K^*(b, 1, s)$ the net expected margin drift under conflict on benchmark b is at least $\eta_R(s)\delta_{b,1}(s) - \xi_{b,1}(s, D)$. If $s \geq s_b^*$, Assumption 7 makes this drift nonnegative until $D_{\max}$, so bounded monotone convergence implies that the margin reaches or remains near saturation. Confidence therefore remains near $\sigma(D_{\max})$, and the signed overconfidence gap at $K_{\max}$ differs from its value at the onset of saturation by $o(1)$, up to the explicit answer-change correction.

If $s < s_b^*$ and $D_K > D_{\text{rec}}$, the same assumption gives net drift at most $-\zeta$. Hence over a sufficiently long post- K^* window, D_K decreases by a nonzero amount before reaching the recovery region. Since σ is strictly increasing, confidence decreases by a nonzero amount. If correction does not reduce accuracy by an equal or larger amount, the signed overconfidence gap strictly decreases, proving the recovery inequality.

Part (d) follows because every step used RLVR-specific sufficient conditions. Helpfulness-tuned models that do not satisfy those conditions are outside the theorem’s scope. $\square$

Remark 2 (What changed relative to the circular proof). *The monotone update is no longer an axiom. It follows from Lemma 2, which itself follows from the primitive fact that self-generated traces are more likely under states that preserve the current answer basin. Likewise, conflict amplification is not assumed as a calibration fact; it follows from a larger self-confirming separation and a lower critical learning rate under conflict.*

Remark 3 (Why ConflictBank and WikiContradict can diverge). *The threshold s_b^* and the critical rate $\bar{\eta}_{b,c}(s)$ are benchmark-conditional. Thus the same RLVR model may satisfy $s \geq s_b^*$ on a benchmark with large self-confirming separation, such as entity-swap conflicts, and $s < s_b^*$ on a benchmark with weaker or more naturalistic contradictions. Equivalently, ConflictBank-style conflicts can have larger $\delta_{b,1}(s)$ and $g_{b,1}(s)$ than WikiContradict-style conflicts, lowering $\bar{\eta}_{b,1}(s)$ and suppressing recovery. Saturation on one benchmark and partial recovery on another are therefore compatible with the theorem rather than contradictory evidence against it.*

5.4 Empirical audits for the CoT assumptions

Answer-stability audit. The proof should be paired with an answer-stability report rather than an unqualified monotonicity claim. For a budget set $\mathcal{K} = \{0, 128, 1024\}$, compute

$$\hat{S}_{\mathcal{K}}^{(b,c,s,\tau)} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\hat{y}_i(K) \text{ is constant over } K \in \mathcal{K}\},$$

and pairwise flip rates

$$\widehat{F}_{K,K'}^{(b,c,s,\tau)} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\widehat{y}_i(K) \neq \widehat{y}_i(K')\}.$$

These quantities instantiate [Lemma 4](#). In the final empirical appendix, report the one-line check “In the ConflictBank conflict subset, the predicted answer remains stable across CoT budgets $\mathcal{K} = \{0, 128, 1024\}$ on $100 \cdot \widehat{S}_{\mathcal{K}}\%$ of examples; pairwise flip rates are $\widehat{F}_{0,128}$, $\widehat{F}_{128,1024}$, and $\widehat{F}_{0,1024}$.” If the flip rates are large, the theorem still applies through the explicit correction term rather than through the stable-answer special case.

Self-confirming likelihood-separation audit. Assumption 4 is directly testable. On a held-out sample of conflict examples, generate a trace Z_i from the model’s current answer basin $a_i = \widehat{y}_i$. Then score the same trace under a prompt or constrained decode that conditions on the current answer a_i and under the strongest competing answer a'_i . Define the length-normalized log-likelihood separation

$$\widehat{\Delta}_i = \frac{1}{|Z_i|} \left\{ \log p_{\theta}(Z_i \mid q_i, c_i, a_i) - \max_{a'_i \neq a_i} \log p_{\theta}(Z_i \mid q_i, c_i, a'_i) \right\}.$$

The primitive separation condition is empirically supported on benchmark b if the lower confidence bound for $n^{-1} \sum_i \widehat{\Delta}_i$ is positive, or equivalently if the sign rate $n^{-1} \sum_i \mathbf{1}\{\widehat{\Delta}_i > 0\}$ is reliably above one half. A compact appendix figure can show the histogram of $\widehat{\Delta}_i$ for the 100-example ConflictBank conflict audit. This check tests the theory’s engine, not its conclusion.

5.5 Eta-tempered answer decoding

The theory also suggests a direct inference-time repair. The key is to temper only the answer-token distribution after the reasoning trace, not the entire reasoning sequence. Given final answer logits ℓ_y after the trace, define

$$p_{\lambda}(y \mid Z_{1:K}, q, c) = \frac{\exp(\lambda \ell_y)}{\sum_{z \in \mathcal{Y}} \exp(\lambda \ell_z)} = \frac{p(y \mid Z_{1:K}, q, c)^{\lambda}}{\sum_{z \in \mathcal{Y}} p(z \mid Z_{1:K}, q, c)^{\lambda}}, \quad 0 \leq \lambda \leq 1. \quad (13)$$

This transformation leaves the reasoning trace intact and scales only the final answer margin.

Corollary 1 (Optimal eta-tempering in a one-dimensional critical-rate model). *Suppose that, conditional on a benchmark b and conflict regime c , the trace contributes a scalar evidence statistic $T_X \geq 0$. The raw RLVR confidence and the truth-calibrated confidence are*

$$C_{\text{raw}}(X) = \sigma(\eta_R(s)T_X), \quad C_{\text{cal}}(X) = \sigma(\bar{\eta}_{b,c}(s)T_X),$$

with $\eta_R(s) > \bar{\eta}_{b,c}(s)$. Answer-token tempering with parameter λ gives

$$C_{\lambda}(X) = \sigma(\lambda \eta_R(s)T_X).$$

Then the pointwise calibration error $|C_{\lambda}(X) - C_{\text{cal}}(X)|$ is minimized at

$$\lambda_{b,c}^*(s) = \frac{\bar{\eta}_{b,c}(s)}{\eta_R(s)} < 1. \quad (14)$$

Consequently, for every $\lambda \in [\lambda_{b,c}^*(s), 1]$ and every $T_X > 0$,

$$C_{\lambda}(X) - C_{\text{cal}}(X) \leq C_{\text{raw}}(X) - C_{\text{cal}}(X),$$

with strict inequality when $\lambda < 1$. For fixed b , if $\bar{\eta}_{b,1}(s) < \bar{\eta}_{b,0}(s)$, then

$$\lambda_{b,1}^*(s) < \lambda_{b,0}^*(s), \quad \frac{\lambda_{b,1}^*(s)}{\lambda_{b,0}^*(s)} = \frac{\bar{\eta}_{b,1}(s)}{\bar{\eta}_{b,0}(s)}.$$

Thus the theory predicts stronger tempering under conflict than without conflict.

Proof. Since σ is strictly increasing, $C_\lambda(X) = C_{\text{cal}}(X)$ if and only if $\lambda\eta_R(s)T_X = \bar{\eta}_{b,c}(s)T_X$. For $T_X > 0$, this gives (14). For $\lambda \geq \lambda_{b,c}^*(s)$, monotonicity of σ gives

$$0 \leq C_\lambda(X) - C_{\text{cal}}(X) \leq C_{\text{raw}}(X) - C_{\text{cal}}(X),$$

with strict inequality for $\lambda < 1$. The conflict/no-conflict comparison follows by fixing b and substituting $c = 1$ and $c = 0$ into (14). $\square$

Remark 4 (Interpreting empirical eta values). *If a validation split on a fixed benchmark estimates $\lambda_{b,1}^* \approx 0.325$ under conflict and $\lambda_{b,0}^* \approx 0.625$ without conflict, the ratio $0.325/0.625 \approx 0.52$ estimates $\bar{\eta}_{b,1}(s)/\bar{\eta}_{b,0}(s)$ in the scalar model. This is precisely the kind of quantitative prediction that distinguishes the theory from a purely descriptive calibration observation.*

6 Paper-safe theorem wording

The following wording is safe for the main paper.

1. **Bayes-optimal arbitration.** The geometric arbitration rule is Bayes-optimal under a log-pool/KL-compromise decision model with posterior mean reliability weight. The correct plug-in regret factor is the log-likelihood-ratio oscillation, not $\|P_X - Q_X\|_1$, unless a linear pool is used.
2. **Fixed trust lower bound.** Fixed trust policies are minimax-suboptimal over benchmark classes containing both temporal-update and misinformation conflicts with separated margins. Signal-adaptive calibration recovers the Bayes rule at the usual plug-in rate.
3. **CoT calibration degradation.** Extended CoT calibration degradation is a conditional consequence of an implicit generalized-Bayes model with endogenous reasoning traces, self-confirming likelihood separation, RLVR-level effective learning rate, benchmark-conditional conflict-lowered critical learning rate, answer-flip correction, and scale-dependent correction-channel capacity. It should not be stated as a universal theorem for all reasoning models; the empirical appendix should report answer-stability and self-confirming-likelihood audits.
4. **Eta-tempered decoding.** If overconfidence is caused by an effective learning rate exceeding the critical rate, answer-token-only eta-tempering is the inference-time repair implied by the theory, with the optimal shrinkage predicted by $\bar{\eta}_{b,c}(s)/\eta_R(s)$.

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